

ENV 4120

Air Pollution and Hazardous Waste Control

Today's Topics

- Air pollution transport and effects
- Expectation
 - Understand and be able to interpret wind rose map
 - Familiar with the wide range of adverse health and environmental effects of air pollution

Transport of air pollution

Pollution transport

- After emitted/formed, pollutants need to be transported to get in contact with “receptor”
 - Human, certain part of eco-system etc..
- Pollution transport depends on meteorology
- Pollutants can be removed or formed during the process
- The distance of transport differ drastically depends on pollutant characteristics

Gone with the wind – Wind Rose

Horizontal mean flow

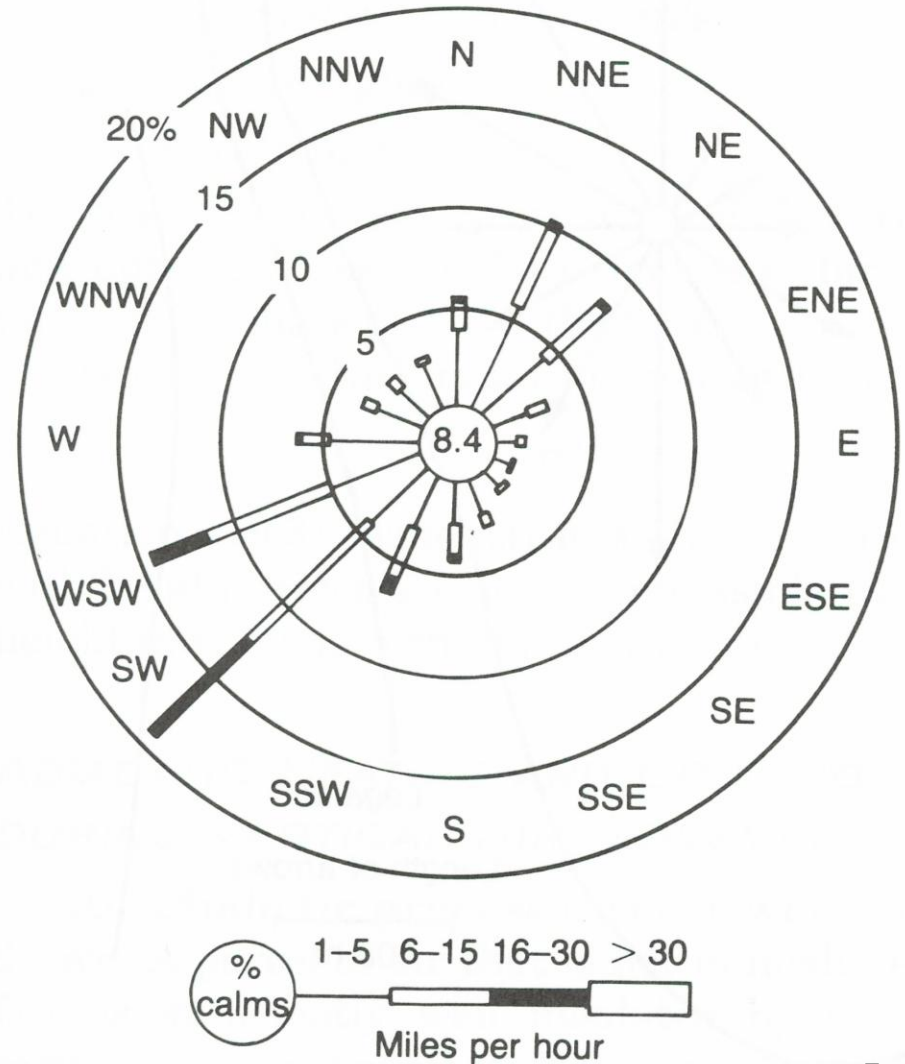
Direction

Where wind is coming from
(angle in degrees clockwise
from the north)

Direction of pollutant transport

Speed

transports pollutants further
dilutes emissions
creates turbulent mixing
bends plumes

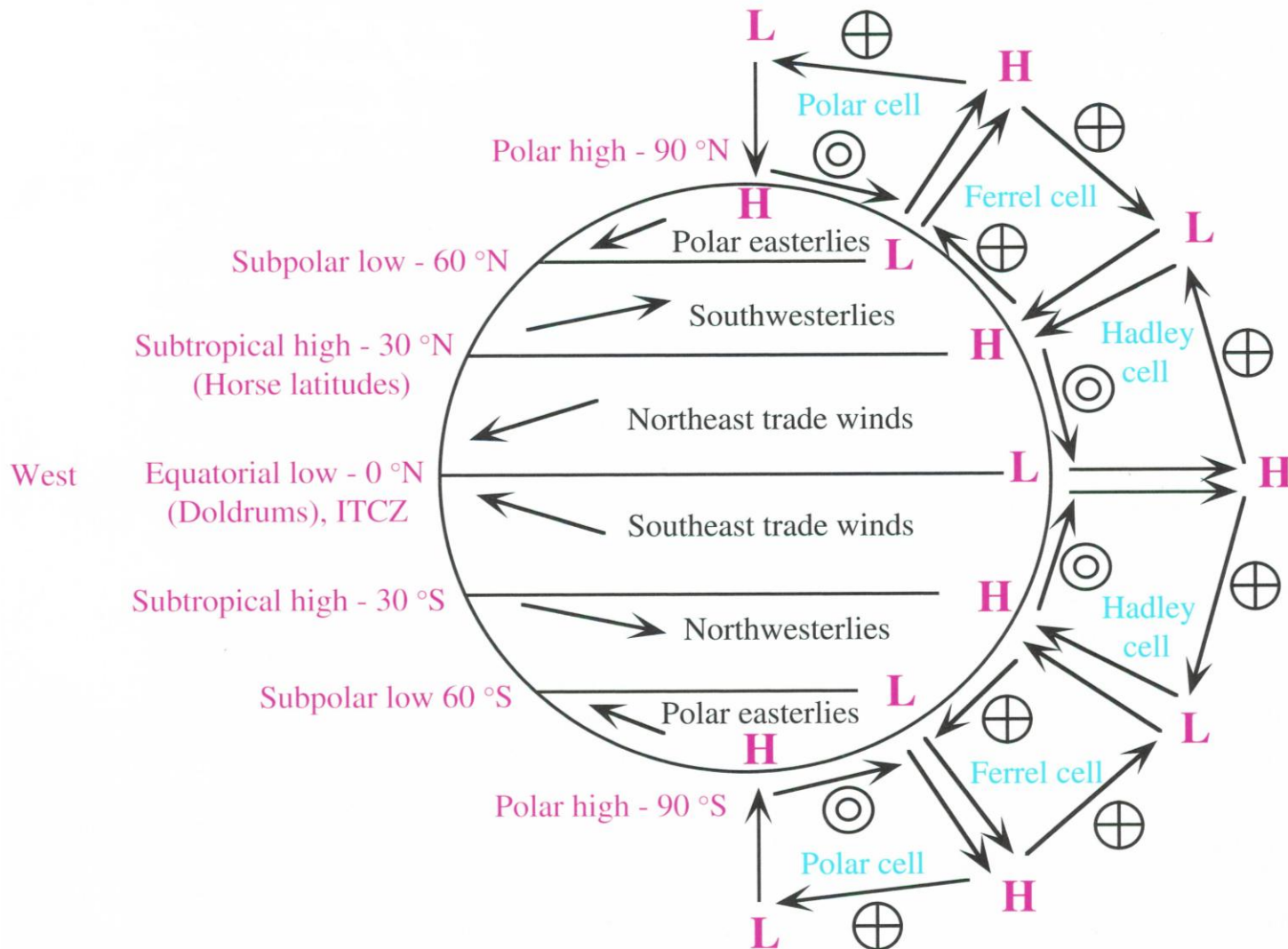


Gone with the wind



Klyuchevskaya Sopka volcano, Russia. 9/30/1994 eruption. Taken by Endeavour
Credit: NASA

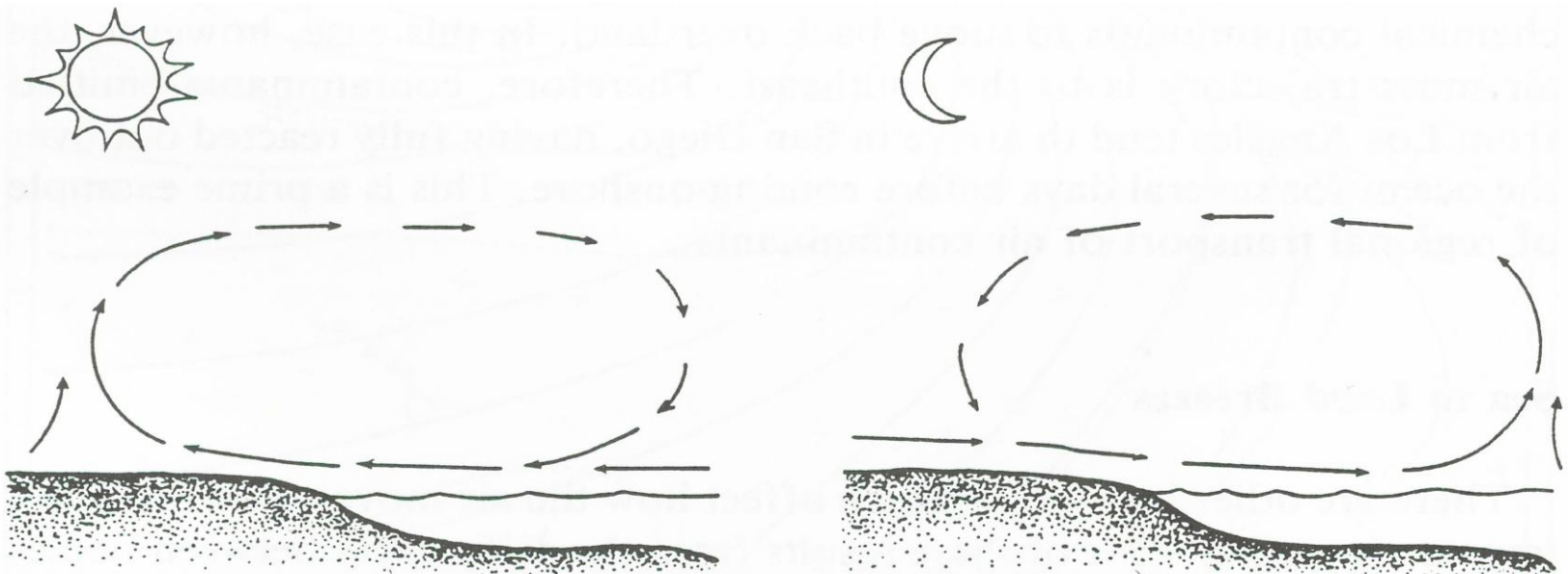
Large Scale Motion: General Circulation pattern



(Jacobson, 2002)

Smaller scale motion

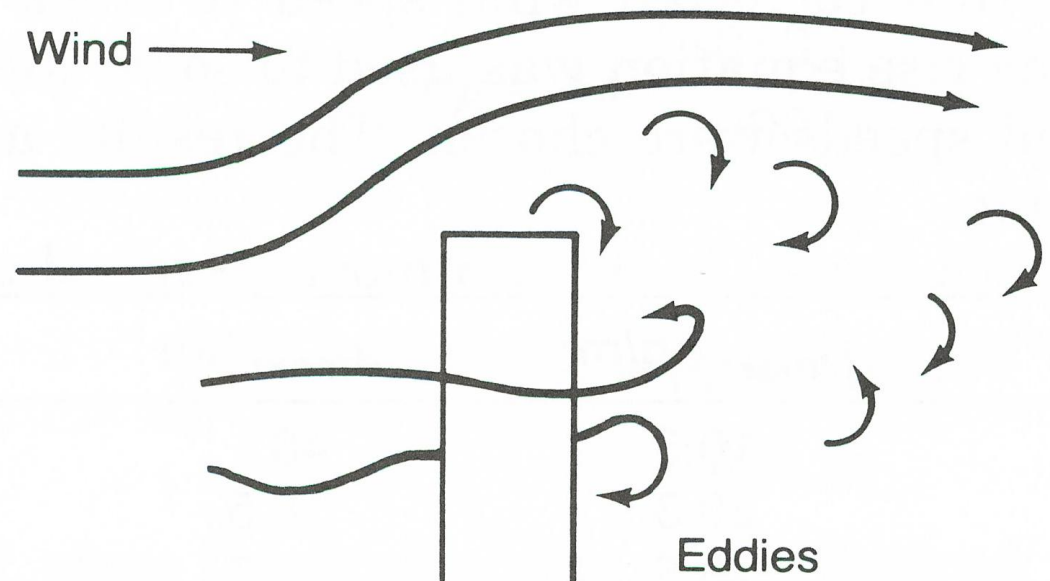
- Impact by variable heating and topography
 - valley and mountain breezes
 - sea and land breezes
 - urban heat island flow



Example of land sea breeze (Griffin, 1994)

Even smaller scale

- Turbulence
 - Random / irregular fluctuations on top of mean flow
 - Circular eddies at smaller scale than mean flow
- Important to
 - mixing
 - dilution
- Pollution hot spot



Effects of air pollution

Air Pollution Health Effects

- Effects on human
 - Numerous adverse health effects, ranging from minor irritation, discomfort, to chest pain, asthma, birth defect, cancer and death
- Effects on the environment
 - Damage property/crops/plants (ozone, acid rain etc..)
 - Visibility impairment
 - Eutrophication (excessive nutrition, e.g. nitrogen)
 - Damage ozone layer → increase harmful UV radiation
 - Numerous others....

Air Pollution Health Effects

Respiratory

Coughing, wheezing,
reduced lung function

Exacerbation of asthma,
COPD

Lung cancer

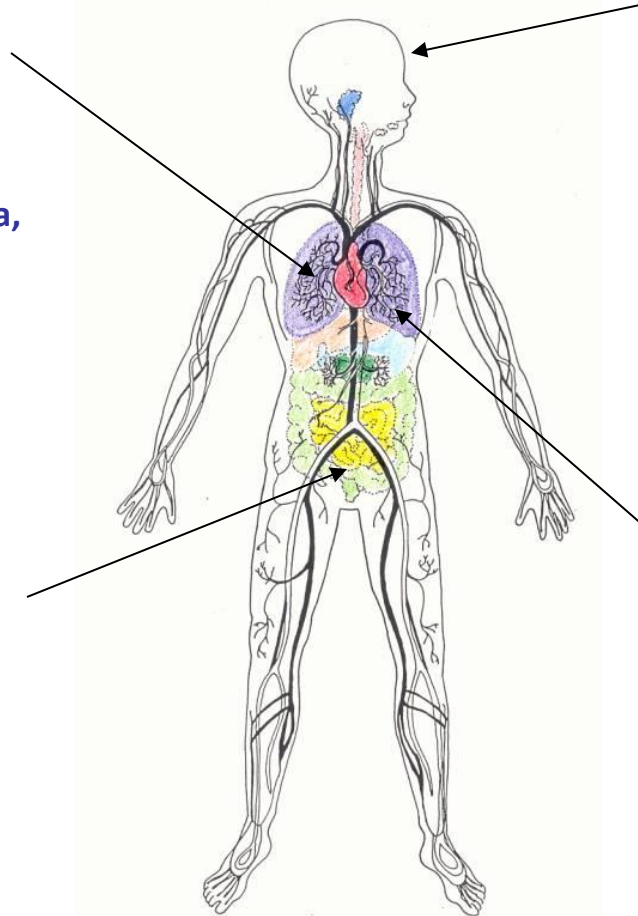
Respiratory mortality

Reproductive

Low birth weight

Preterm births and
intrauterine growth
retardation (?)

Birth defects (?)



Central Nervous

Stroke (?)

Cognitive effects(?)

Cardiovascular

HRV reduction, dysrhythmias

Systemic inflammation

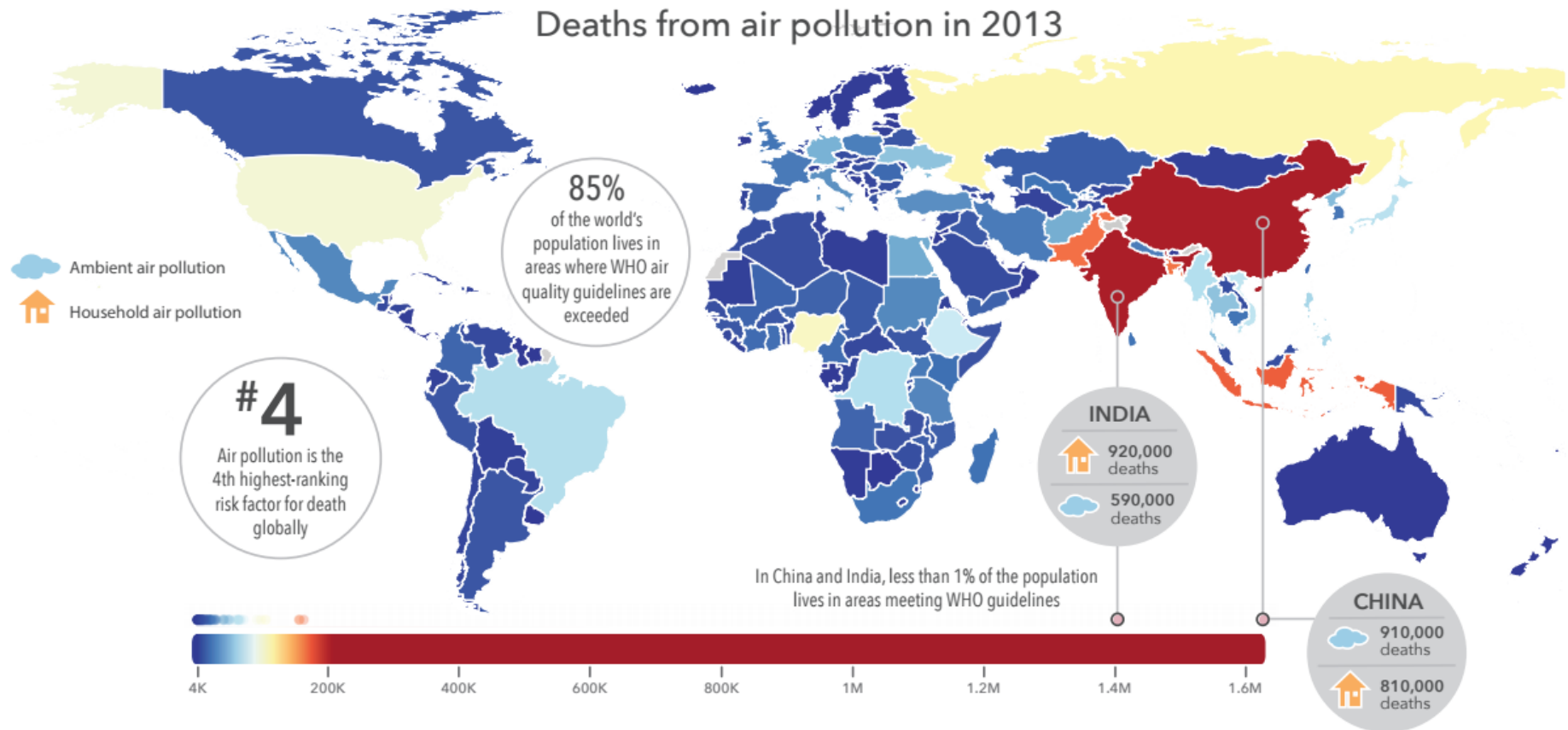
Atherosclerosis

Myocardial infarctions (Heart Attacks)

CV mortality

Air pollution kills

Global Burden of Air Pollution



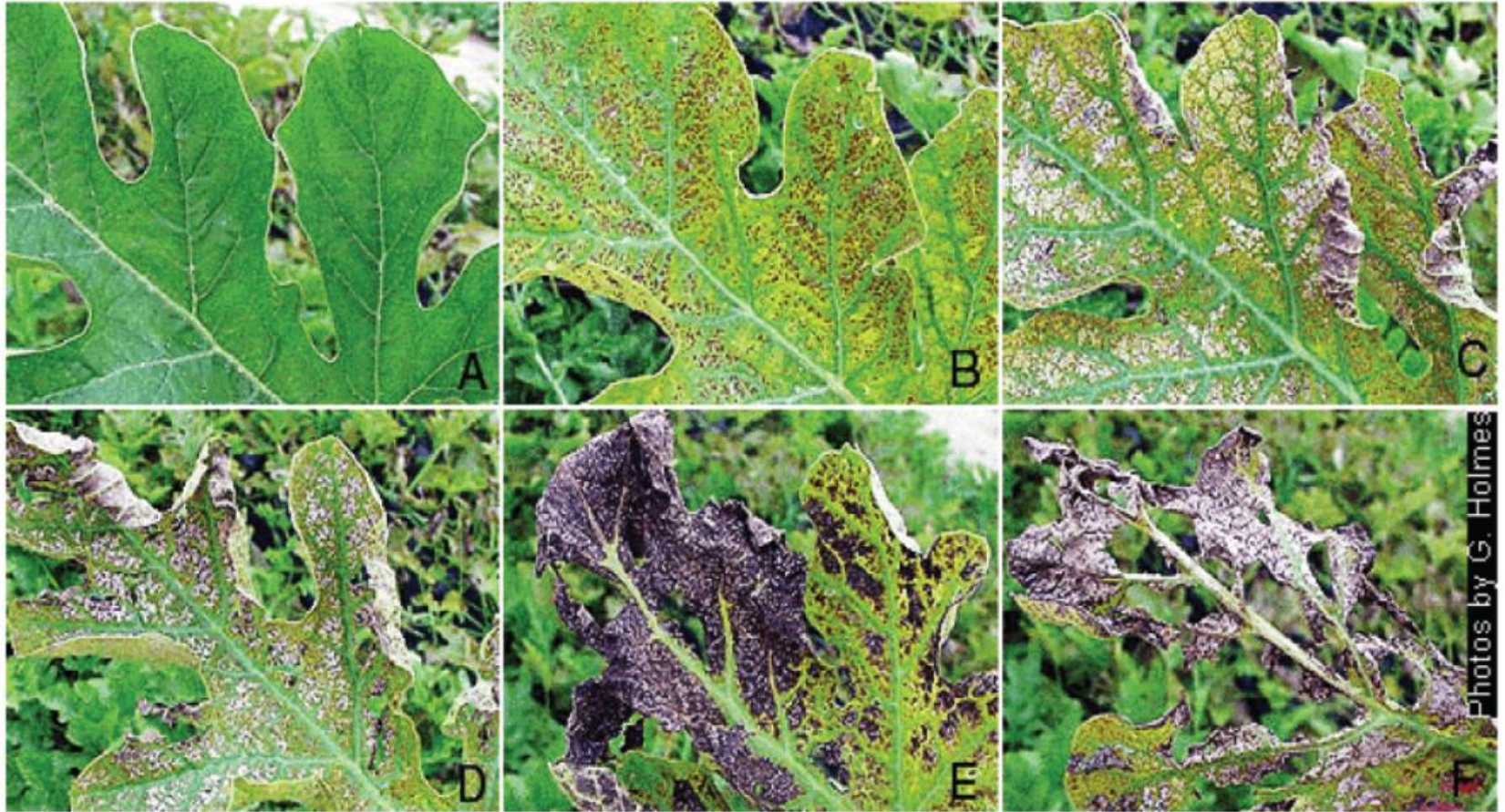
Air pollution was responsible for 5.5 million deaths in 2013

Credit: www.healthdata.org. Global Burden of Disease team

Ozone damaged crops

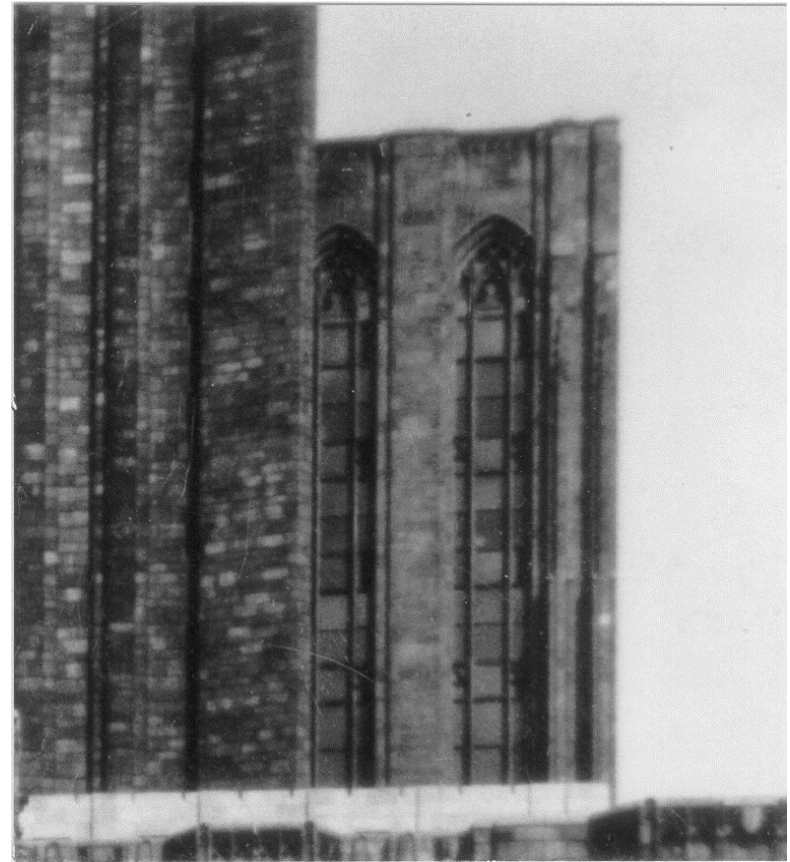
Fig. 2 Progression of ozone damage (A=none to F=severe) on watermelon foliage.

Photo courtesy of G. J. Holmes, NCSU



Photos by G. Holmes

PM deposition on building



Cathedral of Learning, Pittsburgh, 1930 and 1934. (Figure 10.8 Jacobson, 2012) 15

Effect of acid deposition



Sandstone Figure in 1908 and 1968, Westphalia, Germany
(Figure 10.6 Jacobson, 2012)

Effect of acid deposition



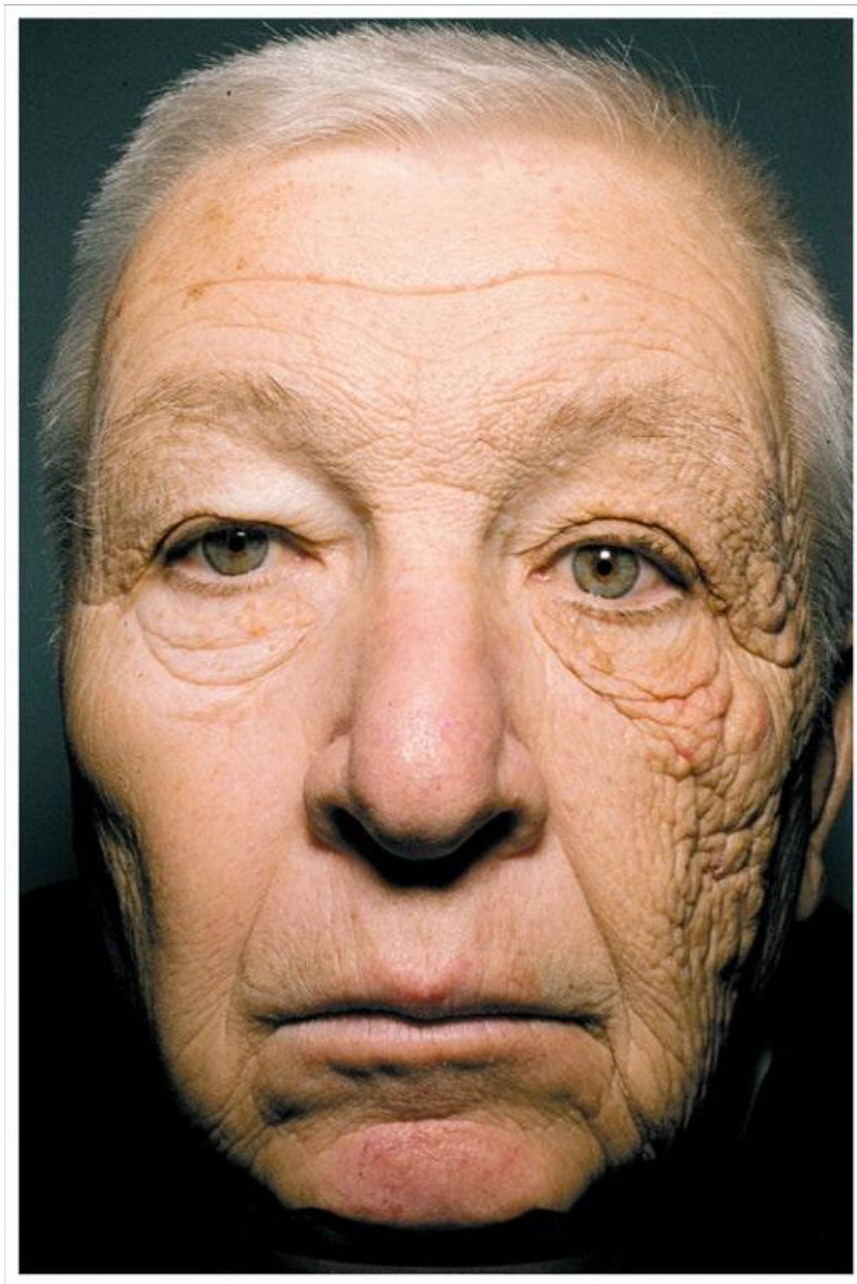
Dead Norway Spruce, Bohemian Forest, Czech Republic, 2005 (www.chmi.cz)

Effects of eutrophication



A boy swims in the algae-filled coastline of Qingdao, China. 2013
(REUTERS/China Daily. <http://zyzc.greatzhonghua.org/archives/14740>)

UV damage skin



Gordon & Brieva. N Engl J Med 2012; 366:e25